

Experiment 04: Input & Output Manual

Create a Simple Login Form using React.js

🎓 Student & Laboratory Credentials

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- **Class / Division:** B.Tech Computer Engineering (Sem VI)
- **Academic Year:** 2026 – 2027
- **Course:** Skill Based Lab - Advanced Web Technology (SBL-AWT)
- **Institution:** Ramrao Adik Institute of Technology (RAIT), D Y Patil Deemed to be University, Nerul, Navi Mumbai
- **Problem Statement Domain:** HealthPulse — Digital Healthcare & Clinical Management Portal
- **Experiment Title:** Create a Simple Login Form using React.js

1. Experiment Aim

To develop a secure, multi-role authentication portal using **React.js** (React 18), utilizing controlled form components, real-time client-side validation, role state transitions, simulated asynchronous authentication latency, and conditional rendering to a protected Clinical Session Dashboard.

2. 📁 Input Specification

2.1 File System Input

File Name	Format	Role & Implementation Responsibility
index.html	React 18 (Babel Standalone)	Single Page Application containing controlled form inputs, multi-role state machine, simulated JWT session generator, and responsive CSS3 theme.
package.json	JSON	Project descriptors, author details, and run configurations.

2.2 Execution Command Input

```
# Navigate to experiment directory
cd /home/ubuntu/SBL-EXPT/Asmit_25CE1051/Expt-04-React-Login-Form

# Open in browser directly
chromium-browser index.html

# Or serve via local server
npm serve .
```

2.3 Controlled State Machine Input Variables

STATE VARIABLE	TYPE	INITIAL VALUE	DESCRIPTION / REACT HOOK ROLE
activeRole	string	'Patient'	Active tab: 'Patient' 'Doctor' 'Admin'
identifier	string	''	Controlled input for Patient Email or 10-digit Phone
password	string	''	Controlled input for Account Secret Key / Password
showPassword	boolean	false	Toggles input type ('password' <-> 'text')
isLoading	boolean	false	Latency simulation flag triggering loading spinner
errorMessage	string	''	Stores client-side validation error message banner
isAuthenticated	boolean	false	Gates conditional rendering of authenticated screen
currentUser	object	null	Persists user identity and session security token

2.4 User Roles & Default Test Profiles (Input Data)

Role Name	Test Username / Identifier	Test Password	Access Scope Granted
Patient	patient@healthpulse.org	Patient@1234	Health Records, OPD Prescriptions, Appointments
Doctor	dr.vidhate@healthpulse.org	Doctor@5678	Patient Consultations Queue, E-Prescriptions
Admin	admin@healthpulse.org	Admin@9999	Server Diagnostics, User Management, Audit Logs

2.5 Validation Input Constraints

- **Identifier:** Must be a valid email (`/^[^\s@]+@[^\s@]+\.[^\s@]+$/`) OR a valid 10-digit phone number (`^[6-9]\d{9}$`).
- **Password:** Must be at least 6 characters in length (`password.length >= 6`).

3. Output Specification

3.1 Graphical User Interface Output Representation (Login State)

```

+-----+
| [HealthPulse Portal]  SBL - Advanced Web Technology (Sem VI)           Student: Asmit (25CE1051) |
+-----+
|
|           ▢ HEALTHPULSE CLINICAL ACCESS
|       Secure Electronic Health Records (EHR) Authentication
|
| +-----+
| | Switch Role:  [ (•) ▢ Patient ]    [ ▢ Doctor ]    [ ♥ Clinical Admin ] |
| +-----+
|
| Patient Email or Registered 10-Digit Mobile:
| [ patient@healthpulse.org ]
|
| Account Password:
| [ ..... ] [ 👁 Show]
|
| [ Fill Patient Demo ]    [ Fill Doctor Demo ]    [ Fill Admin Demo ]
|
| +-----+
| |                               [ ▢ Sign In to Patient Portal ]           |
| +-----+
+-----+

```

3.2 Authenticated Dashboard Output Representation (Conditional Rendering)

Upon successful credential validation and simulated 600ms latency, `isAuthenticated` toggles to `true`. React unmounts the login card and dynamically mounts the **Protected Clinical Session Dashboard**:

```

+-----+
|  HEALTHPULSE CLINICAL SESSION DASHBOARD                               Session ID: hp_jwt_77a91bf2  |
+-----+
| Welcome back, Asmit Jogdand!           Role: [  PATIENT PORTAL ACCESS  ] |
| Active Medical ID: HP-2026-9921         Authenticated At: 2026-10-01 10:25:00 IST |
+-----+
|
| +- Quick Clinical Actions -----+
| | [  View Lab Reports  ]   [  Active Prescriptions  ]   [  Upcoming Consultations  ] |
| |-----+
|
| +- Session Security Telemetry -----+
| | • Protocol:           Encrypted Session (SHA-256 HMAC Token) |
| | • Client IP:          127.0.0.1 (Local Verified Interface) |
| | • Session Timeout:    Valid for 30 minutes |
| |-----+
|
|                                     [  Sign Out of Clinical Session  ]
|
+-----+

```

3.3 Test Case Execution Results

Test Scenario	Injected Input State	React Component & Validation Logic	Rendered UI Output Observed	Status
Empty Inputs	Identifier: "", Pass: ""	Intercepted in <code>handleSubmit()</code>	Red alert: "Please enter your email or phone number"	PASS
Malformed Phone	Identifier: 9820	Regex fails email & 10-digit phone	Red alert: "Please enter a valid email or 10-digit number"	PASS
Short Password	Pass: abc (3 chars)	Evaluates <code>password.length < 6</code>	Red alert: "Password must be at least 6 characters"	PASS
Password Unmask	Click [Show]	State <code>showPassword</code> toggled <code>true</code>	Input type changes to <code>text</code> ; characters visible	PASS
Role Switching	Click [Doctor]	Updates <code>activeRole = 'Doctor'</code>	Active tab pill highlights blue, login button updates text	PASS
1-Click Profile	Click "Fill Patient Demo"	Auto-populates credentials in state	Identifier and password inputs instantly populate	PASS
Loading State	Submit valid credentials	Sets <code>isLoading = true</code>	Button displays spinner: "Authenticating Session..."	PASS
Auth Transition	Latency completes (600ms)	Sets <code>isAuthenticated = true</code>	Login form unmounts; Clinical Dashboard renders	PASS
Session Logout	Click "Sign Out"	Resets <code>isAuthenticated = false</code>	State cleared; returns cleanly to pristine Login Form	PASS

4. Conclusion & Verification

Experiment 04 was completed successfully. React controlled inputs synchronized DOM values directly with state hooks, client validation prevented erroneous payloads, and conditional rendering provided a secure, seamless transition between unauthenticated and protected healthcare portal views.